

An Exploratory Study of Shared Decision-Making (SDM) for Older Adult Patients with Chronic Diseases

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ABSTRACT

Shared decision-making (SDM) is a central feature of clinical practice guidelines, and it is a key way to improve long-term communication and relationships between older adult patients and physicians. Previous studies have found that patients' engagement in decision-making is closely associated with patient satisfaction and with improved treatment outcomes. However, medical decision-making can be complex. Healthcare often requires decisions that are time-sensitive, and when confronted with multiple possible treatments it is necessary to carefully assess the impact of each treatment on health outcomes and to consider potential harms as well as benefits. Effective communication tools can be useful for conducting this assessment in a shared fashion. In the current project, we conducted interviews with 12 participants, including seven older adults with chronic diseases and five clinicians who work with this patient population, with the goal of better understanding both groups' perspectives and needs in the clinical setting. The interviews focused on the potential benefits that can be derived from an SDM-based technological platform, and the findings can be useful in the future design of such a platform.

CCS CONCEPTS

• **Social and professional topics** → **Seniors**; • **Applied computing** → **Health care information systems**; • **Information systems** → **Information extraction**; • **Human-centered computing** → **Empirical studies in HCI**.

KEYWORDS

Clinical decision making, Older adult patients, Chronic diseases, Qualitative method

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1 INTRODUCTION

Patients with low health literacy or knowledge can have difficulty accessing healthcare and understanding their treatment plans. This difficulty has only increased in recent years with the increasing shift toward telehealth, with reduced patient-clinician contact times, and with the impact of the Covid-19 pandemic and other stresses on the healthcare system. The inability to communicate fully and at length with clinicians and to understand the impact of various treatment options can present a barrier to patients' involvement in making a clinical decision that meets their needs and circumstances [4, 11].

Shared decision-making (SDM) is an approach in which clinicians and patients discuss the best available evidence for decision-making and incorporate the patient's preferences into treatment decisions [8, 17, 18]. The SDM process entails the professional providing the patient with information about the decision, presenting various options, and eliciting a discussion about the patient's goals and preferences before ultimately establishing a treatment plan [5, 11]. Previous research has shown that the clinical SDM process can enhance patients' healthcare experience, promote cognitive and emotional gains, create better relationships between patients and clinicians, and improve health outcomes [3, 15, 16]. The use of SDM appears to have positive effects on older adults in particular, including an "improvement in knowledge, increased risk perception, decreased decisional conflict, and an enhancement in participation" [12, 13]. Many older adult patients are struggling with profound life changes, including alterations in their physical, emotional, and social landscapes, which makes them strongly suited for the empowering benefits of the SDM approach. These benefits can also be significant for patients with chronic diseases (which includes many older adults), as they may face complex treatment decisions, with different pros and cons to different treatment options, and may therefore require additional patient support to make informed choices [1, 6].

New technologies provide intriguing modalities for advancing the clinical SDM process, particularly when it comes to enhancing

communication between highly specialized professional healthcare providers and their patients [5]. Many clinicians today are extremely pressed for time, in part due to the staff shortages that remain in the wake of the pandemic, and due to the worsening clinician-to-patient ratios in many healthcare settings. While not a substitute for patient-clinician interactions, SDM-based technological platforms are suitable in some cases to help older adult patients learn about their options in more detail and understand the potential consequences of treatment choices, while easing pressure on clinicians' communication skills [10].

In the current exploratory study, we conducted one-on-one interviews with older adult patients and clinicians, to better understand their needs as related to the SDM process. The main research question evaluated in the interviews was: *How can SDM become more efficient and effective for integrating older adult patients' preferences into treatment decisions?* The answers to this question can potentially lay the groundwork for the future development of technological tools to aid in SDM activities.

2 METHODS

We conducted 12 IRB-approved, semi-structured, one-on-one interviews, with 7 older adult patients (mean age 72) and 5 clinicians (mean years of practice 12), to understand their needs in the clinical decision-making process. Invitations to participate in the study were sent through online platforms and e-mails to assemble a convenience sample. The participants were recruited through online platforms, such as emails and university social media platforms. The older adult patients were recruited through lifelong centers and other nursing homes near the university. A majority of the patient participants were university retirees. The clinicians were recruited through medical school email listservs. We intentionally recruited clinicians and healthcare professionals from diverse chronic disease-related fields. (Tables 1 and 2).

The interviews with older adult patients were divided into 5 question categories, including demographics and basic health information, previous SDM experiences, health literacy, perspectives on health outcomes, and design recommendations for SDM technologies [7]. The interviews with clinician participants were divided into 3 categories: clinical domains and experience, previous SDM experiences, and design recommendations for SDM technologies. All of the interview questions were open-ended. We video-recorded and transcribed the interviews, and used thematic analysis to identify central topics of interest shared by the participants. Each participant received a \$10 gift card as compensation for their time.

By asking the participants to suggest useful features for an SDM tool, we initiated a co-design approach, which involves active engagement and feedback by individuals who are likely to use the technology. In prompting the interviewees to recall their previous SDM experiences, our goal was to encourage them to think about this process and reflect on how it might be improved by technology, thus laying the groundwork for their subsequent recommendations. Our expectation was that this approach would encourage participants to express their opinions and needs more openly and creatively, in contrast to simply asking them to evaluate a list of potential features.

Table 1: Patient participant profiles, including age, gender, and education level.

Patient Participant ID	Age	Gender	Education Level
P1	73	F	Ph.D.
P2	71	F	M.S.
P3	73	F	M.S.
P4	73	F	M.S.
P5	71	F	M.S.
P6	76	F	Ph.D.
P7	68	M	Ph.D.

Table 2: Clinician participant profiles, including practice domain and years of clinical experience.

Clinician Participant ID	Domain	Experience
C1	Psychiatry	6
C2	Cancer	4
C3	Nursing	8
C4	Psychiatry	32
C5	Nursing	10

3 RESULTS

Overall, our interviews indicated that many of the older patients had limited understanding of their treatment options. The patient participants frequently stated that they had difficulty comprehending technical terms related to their conditions, which contributed to their confusion and decreased their ability to engage in collaborative decision-making. The patients indicated that simplifying this technical language and providing fuller explanations would help them to grasp the details of their treatment options and participate more actively in the process. Both the patients and the clinicians indicated that a technological platform that would make it possible to quickly display clear-language explanations of medical concepts would be useful in enhancing the patient-clinician conversation.

Additionally, many of the participants spontaneously suggested that incorporating visual aids, such as graphs and illustrations, would be a useful feature in a technological platform to assist them in processing the information explained by their clinicians. These visual aids cannot by themselves replace the clinicians' verbal explanations, but they would make it easier for patients to visualize the concepts being discussed. This technological potential for visualizing information was mentioned frequently by the clinician participants, but it was also noted by some of the patients, who felt that visual aids would make them more confident in their understanding.

3.1 Additional Patient Responses

The older adult patients who participated in the study indicated that they already employed various technologies to manage their health, including phones, computers, and other smart devices. All of the patient participants reported using telehealth services as one of the means through which they contacted their healthcare providers. A representative quotation in this area was: *"I spend a lot*

of time on my computer, though I'm not an expert. But I don't have any problems [using the information technologies]" (P3).

The patient participants reported that they generally adhered to their health professionals' advice. However, the majority also stated that they had strong opinions about their healthcare and wanted to make their own decisions. For example: *"I try to stay away from surgery as much as possible, and so I research alternative approaches to things. When I found a reasonable alternative approach, I always attempted it first"* (P6). As noted above, our participant sample was well-educated, and in the interview they expressed a good understanding of what constituted good-quality medical information. One stated that: *"I read the New York Times every day, and they have a wellness section. There's a lot of good information in there. Or if I have a question about something like my health or treatment, I know how to look things up and do research"* (P2). Another patient indicated that they were *"leery of a lot of information on the Internet"* (P7) and valued the guidance and assistance of trained clinicians.

Despite this relatively high level of information competence, many of the patient participants felt that they had knowledge-gaps about their treatment plans or what alternatives were available for managing their chronic diseases. This was particularly true when it came to potential harms and adverse effects of treatments, potential drug interactions, and other safety issues. Five out of the seven participants said they felt that their health literacy was relatively weak (P1, P2, P3, P5, P6), and one patient participant expressed uncertain about the effectiveness of the current taking medications (P3).

3.2 Additional Clinician Responses

The majority of the clinicians interviewed confirmed that they viewed SDM as a valuable means of enhancing the management of chronic illnesses among older patients. For example: *"I think the shared decision-making era is coming and like, in some ways, we [clinicians] need to be involved to make sure that it is used appropriately"* (C2). The one clinician participant who did not support the use of SDM stated that the clinicians, in general, have no time for SDM because *"there is a time pressure on each patient visit that really makes it very difficult"* (C5).

Despite their generally positive evaluation of SDM, the other clinicians did express some concerns, mostly related to patients' abilities to assess medical information in an objective and analytical fashion. This included the potential for patients to fixate on treatments that offered potentially transformative solutions, while ignoring low chances of success and potential risks. Several of the clinicians also expressed concern that miscommunications during the SDM process might lead to inaccurate patient expectations, or to clinicians not accurately understanding their patients' goals. In addition, while the patient participants in the current study were all highly receptive to SDM and to the use of associated technology, the clinicians whom we interviewed shared with us that, in some cases, patients are more responsive when clinicians make an authoritative decision and present a single clear plan of action. Finally, clinicians expressed concerns that patients might rely on poor-quality or overgeneralized information resources when understanding their

conditions, and would not understand the best evidence-based approach to addressing their particular circumstances in a treatment plan.

The clinicians indicated that the most useful form of SDM-based technology would be tools that could mitigate these issues by helping patients to understand their suggested treatment options more clearly: *"Obviously, the communication ... is where the technology might be able to make a difference"* (C3). The majority of the clinicians felt that technology could make an important contribution to SDM by assisting in communication efforts (it could *"take load off"* clinicians, according to participant C2), and that it would likely help to improve patient adherence to treatment plans.

3.3 Summary of Technology Recommendations

Combining the interview results from the older adult patients and the clinicians, we identified 10 factors that the participants believed were important to include when presenting information about treatment options to patients. The clinicians gravitated strongly to four factors in particular: medical benefits/effects of treatment, potential risks, detailed information about alternative options, and patient ages at which the treatment was most valuable. The clinicians whom we interviewed impressed upon us the complexity of assessing these factors when tailoring treatment options to an individual patient. It is notable that some of the other factors that multiple patients mentioned as significant, including the cost of the treatment and the travel distance required to access the treatment, were not mentioned by any of the clinicians (Fig. 1).

4 DISCUSSION AND FUTURE WORK

The clinicians and older adult patients whom we interviewed in this study indicated support for the SDM approach and felt that information technology had a valuable role to play in enhancing SDM [2, 16]. The primary value attributed to technology in this process included the ability to improve communication by providing ready-made, clear-language descriptions of medical treatments and their benefits and risks, and to visualize related information through graphs and illustrations [9]. Ten specific factors relevant to presenting treatment-plan information were identified in the interviews (Fig. 1).

It should be noted that the sample size in this study is quite small. The clinician participants were from only three medical fields (psychiatry, cancer, and nursing), and the patient participants skewed heavily toward women and toward highly educated individuals. Additional research needs to be conducted to confirm the findings with a more diverse participant sample. Furthermore, the findings in these interviews involved creative speculation about potential technological benefits, rather than an evaluation of a specific, actually existing platform. Additional robust user-testing will be needed to evaluate the usefulness of a completed product design in clinical SDM processes.

The next stage in this project will be to engage a larger and more diverse sample of chronic disease patients and clinicians to expand our findings about the optimal roles of technology in SDM. We then plan to develop a platform to assist in SDM communication, and conduct usability testing.

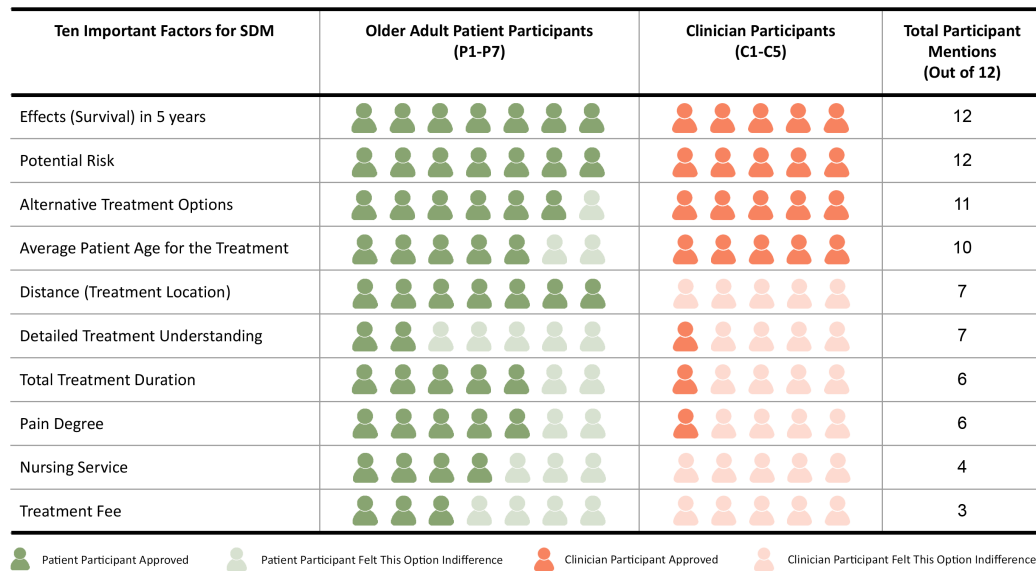


Figure 1: Ten Important Factors in SDM Discussions of Medical Treatments.

Drawing from our literature review and interviews, there are three major obstacles that such a platform will need to overcome:

- **Variability in patient situations and chronic disease characteristics:** Many chronic illnesses are variable diseases, with different stages and grades that require different treatment options. They may also be interconnected with other diseases and other aspects of a patients' life. Thus, for a technological platform to be useful in clinical SDM practice, it will need to incorporate a great deal of nuance in regard to specific information pages that can be brought up for an individual patient, so as to effectively address that patients' situation without confusion.
- **Engaging patients in SDM:** Receptivity to SDM and the desired depth of medical discussions may vary widely for different patients [11]. The trend in current medical practice is shifting strongly toward valuing SDM and patient engagement, but such approaches may not work for all patients. Overall, the technology will need to be situated as one potential tool in the physicians' arsenal, to be judiciously used in situations where patients find it to be desirable and helpful.
- **Time limitations in clinical care:** In developing an SDM-focused application, we will need to be aware that it may be unrealistic to expect many clinicians to have the time required to conduct any type of effective SDM with their patients. The time constraints that the healthcare system places on clinical decisions reduces the possibility of discussing treatment options with patients in depth, even when

demonstrably best medical practices and clinician preferences would point toward the greater use of SDM. Our platform may assist with reducing communication burdens during SDM, but it should be understood as an enhancement, not a substitute, for clinician-patient relationships [14].

5 SUMMARY

In summary, we offered a new approach to understanding the demands of older adult patients and clinicians in clinical decision-making. However, obstacles such as the distrust of the healthcare system, medical knowledge barriers, and urgent time have made it challenging to engage aging populations in the process. Older adult patients may feel neglected or lack self-confidence when directed to the SDM process. The study highlights the need for larger systemic changes to ensure a stronger clinician-patient relationship and how SDM could potentially become a win-win solution. Future research will focus on evaluating optimized solutions to personalize the SDM platform content for specific disease gradations and diverse patient situations, as well as building upon this exploratory study.

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